

PAPER PRESENTATION · THEORY OF CHAIN-OF-THOUGHT

From Sparse Dependence to Sparse Attention

Unveiling how chain-of-thought enhances transformer sample efficiency

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Same model, same parity task: only the training sequence changes

WITHOUT COT

WITH COT

**$2^{\Omega(k)}$
samples**

for any constant-pass SGD
whose memory is $o(nk)$ bits
(Thm 2)

$\tilde{O}(n)$ samples

3 SGD steps reach exact accuracy; attention
becomes one-hot (Thm 3)

No learning

$k = 12$: chance accuracy in
every configuration tried (1–8
layers $\times$ 1–8 heads, 10^4 – 10^6
samples)

$\approx 60,000$

$k = 12$: fresh samples to perfect accuracy, one
pass (App. B.2)

Theory

1-layer model, $o(nk)$ parameters

Experiment

GPT-2-style, Adam, $n = 20$

And expressivity is not the difference: a one-layer model of the same size represents the no-CoT parity exactly (Thm 1).

Why chain-of-thought helps when expressivity is not the bottleneck

EXPLANATION A · PRIOR THEORY

CoT adds computation

Intermediate tokens let a constant-depth transformer run a longer serial computation (Feng et al. 2023; Li et al. 2024; Merrill & Sabharwal 2024).

Explains tasks the model **cannot represent** without CoT.

EXPLANATION B · THIS PAPER

CoT makes learning cheaper

Intermediate tokens change the training signal, so the same function is found from far fewer examples.

Explains failures on tasks the model **can already represent**.

The test: pick a task where A is ruled out (parity is representable without CoT), then ask whether B alone produces a gap, how large, and by what mechanism.

Chain-of-thought now lives in the training data, not only the prompt

Where CoT is used

CoT examples in instruction tuning (Chung et al. 2024), self-generated rationales (STaR), math models trained on CoT corpora (Qwen2-Math), step-level supervision (Lightman et al. 2024).

What theory explained

What a transformer can compute at inference time when it is allowed to write intermediate tokens. Nothing about why training on traces needs less data.

What this paper adds

A provable sample-complexity separation, a mechanism (sparse dependence → sparse attention), and a measurable signature of that mechanism (attention entropy).

If CoT's value is statistical, it says *which* traces help, not just that longer outputs help.

Three practical readings, held as hypotheses

INTERPRETATION

Write traces with sparse dependence

The proof needs each step to depend on one earlier token plus its predecessor. A trace that skips steps re-creates high-order dependence, which is what made no-CoT learning expensive.

EXPERIMENT-BACKED

Watch attention entropy

In multi-pass no-CoT runs, normalized attention entropy keeps falling during the loss plateau, and accuracy jumps when every head is sparse (Fig. 5). A cheap progress signal before the loss moves.

EXPERIMENT-BACKED

Repetition is a costly substitute

$n = 20, k = 6$ without CoT: 10^4 samples $\times$ 1,000 epochs learned; with CoT, 5 epochs. 5×10^4 samples learned where 10^6 at a similar step budget did not.

All three come from a one-task toy setting; slides 29–32 test which parts plausibly transfer to LLM training.

Three obstacles, one per claim

1 No low-order signal without CoT

Under uniform inputs, every set of fewer than k input bits is independent of the parity label. Gradients at initialization carry no information about which bits matter: the statistical-query obstacle (Kearns 1998).

2 A lower bound must cover every trajectory

Good weights exist (Thm 1), so hardness has to come from the learning process. SQ bounds cover full-batch GD, but minibatch SGD can escape them in general (Abbe & Sandon 2020).

3 The upper bound needs finite-sample dynamics through softmax

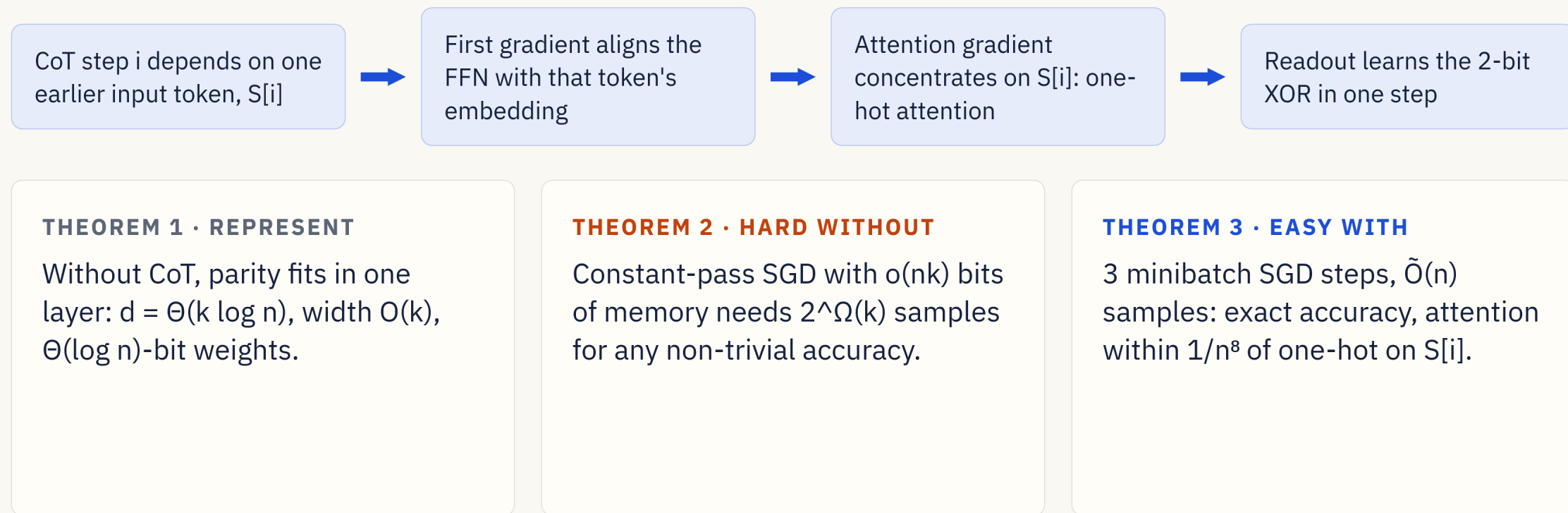
Most transformer-dynamics proofs use population gradients. Here each step uses a minibatch, so martingale noise across all $n \cdot k$ position pairs must be controlled.

Each prior line stops one step short

Line of work	What it established	What it left open (this paper's move)
CoT expressivity: Feng et al. 2023; Li et al. 2024; Merrill & Sabharwal 2024	CoT enlarges what a fixed transformer can represent	Learning at all; tasks already representable
Sub-task decomposition: Wies et al. 2023	Intermediate supervision makes parity poly-sample learnable for RNNs	Kernel (NTK) regime with no feature learning; not transformers
SGD on sparse parity: Barak et al. 2022; Abbe et al. 2023; Kou et al. 2024	$n^{O(k)}$ upper bounds; SQ lower bounds for full-batch GD	An exponential lower bound for minibatch SGD (here: via memory)
Transformer dynamics: Nichani et al. 2024; Wang et al. 2024b	GD learns causal structure and sparse token selection	Population gradients, linear targets, no sample complexity (here: minibatch, XOR target)

Concurrent and not cited in this version: Kim & Suzuki (ICLR 2025) prove a parity-with-CoT result for one-layer transformers by a different route (slide 37).

Three theorems tied together by one mechanism



Experiments: GPT-2-style parity sweeps ($n \leq 100$), multi-pass training without CoT, attention entropy of Qwen2-7B on GSM8K.

Limitations, stated before the details

Scope	What the results assume
Task	Parity with one fixed secret set; each CoT token depends on exactly one earlier input token (the authors' own stated limitation)
Model	One layer, frozen random embeddings, merged query-key matrix, concatenated residual, fixed output layer, no normalization
Training	Three SGD steps with hand-set, stage-specific step sizes; hinge loss; exact CoT supervision available at no cost
Lower bound	Holds only while parameters $\times$ precision is $o(nk)$ bits and passes over data are constant
Regime	Asymptotic: k between $n/\log^5(n/\delta)$ and $n/\log^4(n/\delta)$; polylog factors are enormous (slide 20)
Real data	GSM8K evidence is correlational: attention entropy, no intervention

What the paper contributes, and where each part is covered

Contribution	Result type	Slides
Formal setup: parity with and without CoT on a simplified transformer	Preliminaries	11–15
Thm 1: no-CoT parity representable by one small layer	Representation	16, 22
Thm 2: $2^{\Omega(k)}$ samples without CoT	Learnability lower bound	17, 23
Thm 3: $\tilde{O}(n)$ samples with CoT; one-hot attention	Optimization + generalization	18, 24–27
Experiments: synthetic sweeps, multi-pass, GSM8K	Empirical	33–35
Our analysis: what carries the result, realism, adjacent work, one decisive test	Critique	28–32, 36–40

Widom's guideline 1: the contribution is fully stated by the first quarter (slide 10 of 40). From here on, technical depth.

Running example: $n = 5$, $k = 3$, $S = \{1, 2, 4\}$



$$b[7] = b[1] \oplus b[2] \oplus b[4] = 0$$

One target that depends on three inputs jointly; no single input is correlated with it.

$$b[7] = b[6] \oplus b[1] = 0$$

$$b[8] = b[7] \oplus b[2] = 1$$

$$b[9] = b[8] \oplus b[4] = 0$$

Each target depends on its predecessor and one input.
Ideal attention: $6 \rightarrow 1$, $7 \rightarrow 2$, $8 \rightarrow 4$.

The data-generating process

Def. 1 · parity (n, k) without CoT

$b[i] \sim \text{Unif}\{0,1\}$ i.i.d., $i = 1 \dots n$

$b[n+1] = 0$ (EOS)

$b[n+2] = \oplus_{j \in S} b[j]$

Length $n + 2$; one supervised position.

Def. 2 · parity (n, k) with CoT

$b[1 \dots n+1]$ as in Def. 1

$b[i+1] = b[i] \oplus b[\tilde{S}[i]],$

$i = n+1, \dots, n+k$

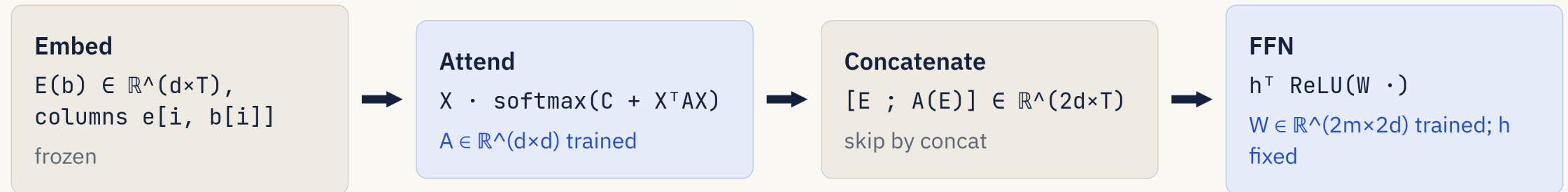
Length $n + k + 1$; $\tilde{S}$ is any fixed ordering of S ; k supervised positions.

Fixed: the secret set S , $|S| = k$, shared by training and test. k sets the hardness.

CoT content: running prefix parities. The last CoT token equals the no-CoT answer.

Sampling: each training example is a fresh draw of b ; the sample size counts these draws.

The hypothesis class: a simplified one-layer transformer



Component	Definition in the paper (Defs. 3–6)
Embeddings	For each position i and bit b , $e[i, b] \sim \text{Unif}\{\pm 1/\sqrt{d}\}^d$, sampled once and frozen: near-orthogonal w.h.p.
Attention	One head; the query-key product merged into a full matrix A ; value and output maps are the identity; causal mask C
Residual and norm	DenseNet-style concatenation instead of addition; no LayerNorm or BatchNorm
Output	Scalar logit per position; $h = +1/(2m)$ on half the neurons and $-1/(2m)$ on the other half

Loss, training rule and evaluation criterion

Element	Specification
Loss	Hinge $\ell(\hat{y}, y) = \max\{(-1)^y \cdot \hat{y} + 1, 0\}$. No CoT: position $n+1$ only. CoT: $\sum$ over $i = n+1 \dots n+k$ of $\ell(T(b)[i], b[i+1])$, with the true prefix as input (teacher forcing).
Rule, Thm 2	Any random init, constant depth; q -pass minibatch SGD, q constant; total state $O((d^2 + dm) \cdot \log n)$ bits.
Rule, Thm 3	One pass, 3 SGD steps on fresh batches of $B = C_2 \cdot n \cdot \log^{20}(n/\delta)$; $\eta_0 = \eta_1 = m\epsilon\sqrt{B} / (100 \log(n/\delta))$, $\eta_2 = 4k\epsilon/3$; trains A and W .
Init, Thm 3	$A = 0$ (uniform attention); W 's attention block = 0; $W[r, 1:d] = \sum v[r,i,b] \cdot e[i,b]$ over CoT positions, $v \sim \text{Unif}\{\pm\epsilon\}$; $d = k \cdot \log^{1.1}(n/\delta)$, $m = 2k$.
Evaluation	Thms 1, 3: correct sign at every supervised position for every input (exact, not average). Thm 2: accuracy $> 50\%$ with probability $> 50\%$. Experiments: samples seen when validation accuracy first hits 100% on 2,048 held-out inputs.

A learnability separation, with representation as the control

Category	Thm 1	Thm 2	Thm 3
Representation	Primary: weights exist	none	Secondary: the learned weights
Computational complexity	none	Secondary: memory–sample trade-off	none
Trainability	none	none	Subsumed by optimization
Optimization	none	none	Primary: explicit SGD path
Learnability	none	Primary: lower bound	Primary: upper bound
Generalization	none	none	Secondary: exact on all inputs
Certification	none	none	none

The lower bound is **not statistical**: with $O(n^2)$ bits of memory, Gaussian elimination over $GF(2)$ recovers S from $O(n)$ samples. The gap is about learners whose state is small.

Theorem 1: parity is easy to represent without CoT

For any $\delta < 0.1$ and all sufficiently large n , if $k \in [n/\log^5(n/\delta), n/\log^4(n/\delta)]$, then with probability $\geq 1 - \delta$ over the random embeddings **there exist** weights with $d = \Theta(k \log(n/\delta))$, width $2m = O(k)$ and $\Theta(\log n)$ -bit precision, so that $d^2 + dm = o(nk / \log n)$, and **for every** input b in the support of p_{NoCoT} the sign of $T(b)[n+1]$ encodes $b[n+2]$ correctly.

Quantifiers

Existential over weights; high probability over the frozen embeddings; universal over inputs.

Budget

$o(nk / \log n)$ parameters: inside the budget where Thm 2 forbids learning.

Role

Removes expressivity as an explanation for the gap that follows.

Theorem 2: without CoT, learning is exponentially expensive

For **any** randomly initialized simplified transformer of constant depth with $d^2 + dm = o(nk / \log n)$, and **any** constant number of passes q : if it is trained by q -pass SGD using $O((d^2 + dm) \cdot \log n)$ bits of memory, then reaching accuracy $a > 50\%$ with probability $p > 50\%$ on parity (n, k) without CoT requires $2^{\Omega(k)}$ samples.

Quantifiers

Universal over initialization, step sizes and batch sizes: anything that runs within that memory.

Dependence

Exponent in k alone. In the stated regime $k \approx n / \text{polylog } n$, so the bound is $2^{\Omega(n / \text{polylog } n)}$.

Silent on

Many-pass training, which succeeds in the paper's own runs, and any model with $\geq nk$ bits of state.

Theorem 3: with CoT, three SGD steps suffice

For any constant $\delta \in (0, 1)$ and $k \in [n/\log^5(n/\delta), n/\log^4(n/\delta)]$, with probability $1 - \delta$, the one-layer model of Assumption 1 ($o(nk)$ parameters), trained for a constant number of minibatch SGD steps on $\tilde{O}(n)$ samples with the Assumption 2 hyperparameters, predicts **every** CoT token correctly for **every** $b \sim p_{\text{CoT}}$. Moreover, for all $j \leq i$,

$| \text{softmax}(C + E^\top A E)[j, i] - 1(j = \bar{S}[i]) | < 1/n^8$
and both hold with $\Theta(\log n)$ -bit weights and activations.

Exact rollout

Correct on every true prefix implies free-running generation is also exact: no error compounding.

Interpretability is proved

The learned attention reads off S and its order: a theorem here, not an observation.

Finite samples

Minibatch gradients, not population gradients: among the first such analyses for transformers.

POINT 3 · DEPENDENCE ON THE PROBLEM PARAMETERS

How the bounds depend on each parameter

Parameter	No CoT (Thms 1, 2)	With CoT (Thm 3)
Samples	$2^{\Omega(k)}$ necessary	$3 \cdot C_2 \cdot n \cdot \log^{20}(n/\delta) = \tilde{O}(n)$ sufficient
Dimension d	$\Theta(k \log(n/\delta))$ represents; lower bound needs $d^2 + dm = o(nk / \log n)$	$k \cdot \log^{1.1}(n/\delta)$
Width $2m$	$O(k)$	$4k$
Depth	one layer (Thm 1); any constant (Thm 2)	one layer
Precision	$\Theta(\log n)$ bits per number	$\Theta(\log n)$ bits, weights and activations
Passes	any constant q	one pass, three steps
Conditioning	random $\pm 1/\sqrt{d}$ embeddings; secret block has $\lambda_{\min}(MM^T) > 1/12$	pairwise $ \langle e, e' \rangle \leq \sqrt{(2 \log(8nd/\delta))/d}$
Task size k	sets the exponent	enters only through the regime $k \approx n/\text{polylog } n$

Fig. 3a ($n = 100$): CoT samples grow about linearly in k ; the theorem cannot speak to k at fixed n

What "sufficiently large n " and $\tilde{O}(n)$ hide

[154, 3,560]

The window for k at $n = 10^9$, $\delta = 0.1$.
At $n = 10^6$ it is $[0.9, 14.8]$.

1.8×10^{27}

$\log^{20}(n/\delta)$ at $n = 10^9$: the polylog in
each batch $B = C_2 \cdot n \cdot \log^{20}(n/\delta)$, $C_2 =$
 $1.28 \times 10^7/\epsilon^2$.

$\ln n \gtrsim 10^{46}$

Lemma 10 needs $\sqrt{(2k \cdot \log(100Bm/\delta))/d} < 0.01$; with $d = k \cdot \log^{1.1}(n/\delta)$
this decays like $2 \cdot \log^{-0.05}(n/\delta)$
and is ≈ 2.7 at $n = 10^9$.

The separation is a statement about exponents as $n \rightarrow \infty$. The $n \leq 100$ experiments illustrate the phenomenon; they are not instances of the theorem.

Three theorems, three different proof techniques

THM 1 · CONSTRUCTION

Explicit weights + random-matrix concentration

Dual vectors to the secret embeddings via a pseudo-inverse; Hoeffding bounds for the other positions; a ReLU zigzag turns a count into a parity.

Slide 22

THM 2 · REDUCTION

SGD is a small-memory streaming algorithm

Simulate q -pass SGD as a q -pass branching program with $O(P \log n)$ bits, then invoke Lyu–Tal–Wu–Yang (2023).

Slide 23

THM 3 · DYNAMICS

Three SGD phases, each with one job

Frozen activation patterns, large stage-wise steps, and martingale bounds on six batch statistics (Table 1).

Slides 24–27

Only Theorem 3 analyses training. Theorem 2 never looks at a gradient.

POINT 4 · THEOREM 1 MECHANISM

Count the secret bits, then fold the count into a parity

1 · ATTEND

$$A = 40 \cdot \log n \cdot V e[n+1, 0]^\top$$

Logit $40 \log n$ on S , $\leq 20 \log n$ elsewhere: weight $\approx 1/k$ on each secret position.



2 · READ

$$\langle U, \text{attn} \rangle \approx (1/k) \cdot \sum b[j]$$

Dual vectors U pick out the bit value: the fraction of ones among the secret bits.



3 · FOLD



$2m = O(k)$ ReLUs with alternating slopes map count c to $(-1)^{c+1}$.

This attention is **k-sparse** (uniform over S), not one-hot. Sparse attention exists without CoT; what CoT changes is whether gradient descent can find it.

A small learner cannot keep enough equations

1 · IMPORTED THEOREM

A q -pass learner of k -sparse parities with $o(nk)$ bits of memory needs $2^{\Omega(k)}$ samples (Lyu–Tal–Wu–Yang 2023, after Raz 2016 and Kol–Raz–Tal 2017).



2 · SIMULATION

q -pass SGD on P parameters at $\Theta(\log n)$ bits is such a learner: $O(P \log n)$ bits hold weights, optimizer state and one batch's gradient sum. The frozen embeddings would cost $O(nd)$ numbers, so regenerate them from a fixed seed.



$$P = d^2 + dm = o(nk / \log n) \Rightarrow 2^{\Omega(k)} \text{ samples}$$

Intuition

Each sample is one linear equation over $\text{GF}(2)$ in the indicator of S . Solving needs about n of them; holding them takes about nk bits. A smaller learner keeps forgetting.

Uses only

Memory size, streaming access, constant passes. Not attention, the loss, the initialization or step sizes: Adam or any other architecture of that size is covered too.

Three SGD steps, three jobs

STEP 1 · η_0 LARGE

Point the FFN at $\tilde{S}[i]$

A stays 0: its gradient vanishes while W's attention block is 0. That block moves toward $e[\tilde{S}[i], \cdot]$, separately for each current bit.



STEP 2 · $\eta_1 = \eta_0$

Make attention one-hot

A now receives gradient through the softmax. Its logit on $(\tilde{S}[i], i)$ beats every other by more than $20 \log n$.



STEP 3 · η_2 SMALL

Learn the readout

Attention returns $e[\tilde{S}[i], b[\tilde{S}[i]]]$; W learns the 2-bit XOR. Output $\approx (\epsilon/3) \cdot (-1)^{(b[i+1]+1)}$.

Invariant through all steps	Why it holds
Activation patterns frozen	Neuron r fires at position i iff $v[r, i, b[i]] > 0$; updates move pre-activations by $< 3\epsilon/40$
Step sizes stage-specific	$\eta_0 = \eta_1 = m\epsilon\sqrt{B} / (100 \log(n/\delta)) \gg \eta_2 = 4k\epsilon/3$, with $\eta_0\eta_1 / (256mn^2) > 40 \log n$

Step 1: CoT creates a first-order signal

THE IDENTITY THE WHOLE PROOF RESTS ON

$$(-1)^{b[i+1]} = (-1)^{b[i]} \cdot (-1)^{b[\bar{S}[i]]}$$

Given the current token, the target depends on one input bit: a 1-sparse search over n positions.

With CoT

$$\begin{aligned} &E[(-1)^{b[i+1]} \cdot 1(b[j]=b_1, b[i]=b_2)] \\ &= \pm 1/4 \text{ if } j = \bar{S}[i], \text{ else } 0 \end{aligned}$$

Per batch: signal $B/4$ at the secret position against noise $O(\sqrt{B \log(nB/\delta)})$ elsewhere.

Without CoT

$$\begin{aligned} &E[(-1)^{b[n+2]} \cdot g(b_T)] = 0 \\ &\text{for every set } T \text{ with } |T| < k \end{aligned}$$

The first gradient is mean-zero noise of size about $1/\sqrt{B}$: nothing to amplify.

How the FFN conditions on $b[i]$: at initialization each neuron's on/off state at position i is set by $v[r, i, b[i]]$, so the update is split by current bit for free.

Step 2: the attention gradient lands on the secret index

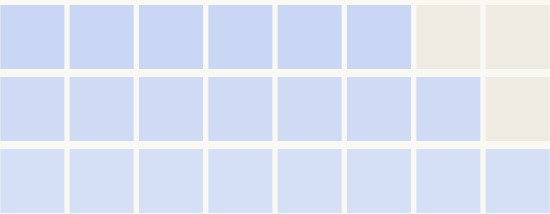
$$d\ell/dA = (-1)^{(b[i+1]+1)} \cdot \sum_j \langle \kappa, e[j, b[j]] \rangle \cdot \partial \text{softmax}[j, i] / \partial A$$

κ is the FFN's read-out direction after step 1. Its overlap with $e[j, \cdot]$ is about $\eta_0 \bar{\alpha} / (8Bmi)$ at $j = \tilde{S}[i]$ and tiny elsewhere, so the update concentrates on the pair $(\tilde{S}[i], i)$.

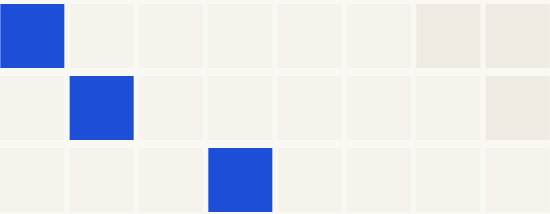
Logit after step 2	Bound (Lemma 21)
On target, $j = \tilde{S}[i]$	$\geq \eta_0 \eta_1 / (256mn^2) > 40 \log n$
Off target	$\leq \eta_0 \eta_1 / (512mn^2)$

Softmax margin (Lemma 32): a gap above $20 \log n$ leaves under $1/n^{10}$ of the mass elsewhere.

RUNNING EXAMPLE · QUERIES 6, 7, 8



A = 0: uniform over the causal prefix



After step 2: 6 → 1, 7 → 2, 8 → 4 (keys 1–8, left to right)

Step 3: XOR becomes linear once the current bit gates the neurons

With one-hot attention the FFN sees $[e[i, b[i]] ; e[\bar{S}[i], b[\bar{S}[i]]]]$. The first half decides **which** neurons fire (the frozen pattern); within that set the output is **linear** in the second half.

XOR of two bits is "copy $b[\bar{S}[i]]$ " when $b[i] = 0$ and "negate it" when $b[i] = 1$: linear inside each gate. Neurons with $v[r,i,0] \cdot v[r,i,1] < 0$ fire for exactly one value of $b[i]$ and carry the signal.

$$T(b)[i] \approx (\epsilon/3) \cdot (-1)^{(b[i] + b[\bar{S}[i]] + 1)}$$

$b[i]$	$b[\bar{S}[i]]$	$b[i+1]$
0	0	0
0	1	1
1	0	1
1	1	0

Unshaded rows: the gate for $b[i] = 0$ (copy). Shaded rows: the gate for $b[i] = 1$ (negate).

Hinge loss: outputs stay within $\pm\epsilon/3 \leq 1$, where the hinge is linear, so the linear-loss analysis transfers (Lemma 27).

Two reasons no-CoT learning fails; the paper proves one

INFORMAL · PROOF SKETCH, STEP 1

Correlation decay

No set of fewer than k bits predicts the label, so early gradients carry no signal. Holds for models of **any** size; gradient methods then pay $n^{\Theta(k)}$ -type costs (Barak et al. 2022).

Status here: intuition only. Kim & Suzuki (2025) prove an SQ-type bound of this kind in their setup.

FORMAL · THEOREM 2

Memory starvation

A learner holding $o(nk)$ bits cannot retain enough equations, whatever its gradients do. Holds only for **small** models trained with constant passes.

Status here: proved.

The experiments use about 10^5 times more parameters per layer than the $o(nk / \log n)$ budget. Whatever slows them down is the left mechanism, which this paper does not prove.

Bounded memory and constant passes carry the separation

Theorem 2 needs learner state of $o(nk)$ bits **and** $O(1)$ passes. Remove either and the exponential disappears.

More memory

Gaussian elimination with $O(n^2)$ bits learns no-CoT parity from $O(n)$ samples.
Abbe & Sandon (2020): SGD on a size- n^2 net learns parities in polynomial time.

More passes

The paper's Fig. 4: 10^4 samples, about 1% of the 2^{20} inputs, $\times 1,000$ epochs learn $n = 20$, $k = 6$ in almost every configuration.

CoT side: sparse dependence

Each CoT step depends on exactly one earlier input bit plus its predecessor. That is what makes the step-1 signal first-order.

So "exponential sample complexity" here means a memory–sample trade-off for streaming learners, not a statistical barrier.

One model must satisfy both theorems at once



Margins that are powers of $\log n$ are why the constants only settle at astronomical n (slide 20).

How closely the formal model matches an LLM

Component	This paper	Modern LLM	Load on the result
Embeddings	frozen random $\pm 1/\sqrt{d}$	learned; rotary positions	High: near-orthogonality used
Attention	1 head, full A, identity V/O	many heads, low-rank QK, V, O	Medium: a one-hot pointer fits
FFN	1 ReLU layer, output fixed	trained GeLU/SwiGLU blocks	High: frozen gates make XOR linear
Optimizer	3 SGD steps, huge step sizes	Adam, $\sim 10^5$ steps, small LR	High: one-step feature learning
Data	one task, exact CoT, fresh	task mixture, noisy traces	High: sparse dependence assumed
Size vs task	$o(nk / \log n)$ parameters	far above nk for any one task	Decisive for Theorem 2

Plausibly transfers: attention as a learned pointer to the one relevant earlier token, as copy-style heads do in LLMs.

Does not transfer: the memory-starved lower bound, since LLMs hold far more than nk bits.

The experiments sit far outside Theorem 2's regime

6.5 M

parameters in one GPT-2-style layer used in the experiments (hidden 720, MLP 3,072)

35

$nk / \ln n$ at $n = 30$, $k = 4$: the order of Theorem 2's parameter budget, constants aside

1.8×10^5

times over budget, per layer, before counting 32-bit precision

A 7B model in bf16 holds 1.1×10^{11} bits. To be memory-starved on one parity-like task, $nk / \ln n$ must exceed 7×10^9 : a 10^6 -token input whose answer depends jointly on about 10^5 positions.

Yet one-pass training without CoT still grows exponentially in k (Figs. 1–2). Theorem 2 cannot explain that; correlation decay is the likely driver (interpretation).

Synthetic experiments: what was measured and what it shows

Experiment	Result	Relation to the theorems
Fig. 2 · $n = 30$, $k = 1-4$, $1-4$ layers $\times$ $1-4$ heads, one pass	No CoT: about exponential in k for every architecture; CoT far lower	Same shape as Thms 2–3, far outside their regime
Fig. 3a · $n = 100$, one layer and head, $k = 20-100$	No CoT exceeds 10^7 samples once $k \geq 4$; CoT learns every k , about linearly in k	k -scaling at fixed n is not covered by Thm 3
Fig. 3b · $n = 40$, $k = 20$, one layer and head	CoT token i attends to $\tilde{S}[i]$, little mass elsewhere	Direct check of Thm 3's attention claim
Figs. 4–5 · $n = 20$, $k = 6$, no CoT, many epochs	10^4 samples $\times$ 1,000 epochs learns; 10^6 at a fixed step budget fails; success follows the entropy drop	Outside Thm 2 (many passes); consistent with its pass dependence

Widom's test: the sweeps vary k and architecture. They never vary n at fixed k , or model size relative to nk , the one variable Theorem 2 depends on.

GSM8K attention entropy: suggestive, with confounds

EXPERIMENT

Qwen2-7B reads each GSM8K test question followed by either the worked solution or "The answer is X."; Qwen2-Math-7B reads the worked solution.

$$\text{Ent} = \min_{i \geq 2} \text{over } i \text{ of } H(P[i]) / \log i$$

Per head, averaged over the test set. Lower with the worked solution; lower still for Qwen2-Math.

OUR CRITIQUE

1 · The metric favours longer text

The question prefix is identical and attention is causal, so both minima share it; the CoT version then minimizes over a much longer continuation. Needs a length-matched control.

2 · Qwen2-Math differs in more than CoT

Continued training on a math corpus could lower entropy by itself.

3 · Nothing is manipulated

Sparsity is measured, never intervened on, so "key factor" is a causal claim the data cannot carry.

Proved, measured, or interpreted

THEOREM

Simplified model, asymptotic regime only.

- No-CoT parity fits in one layer (Thm 1)
- No CoT: $2^{\Omega(k)}$ samples for small-memory, constant-pass SGD (Thm 2)
- CoT: $\tilde{O}(n)$ samples, exact accuracy, one-hot attention (Thm 3)

EXPERIMENT

GPT-2-style at $n \leq 100$; Qwen2 on GSM8K.

- One pass: no-CoT cost about exponential in k , CoT about linear (Figs. 1–3)
- CoT attention reads off S (Fig. 3b)
- Many passes learn without CoT; success tracks entropy (Figs. 4–5)
- Lower entropy on CoT text (Fig. 6)

INTERPRETATION

Claimed or implied; not established.

- Sparse attention is "a key factor" of CoT's gain
- Real CoT data has sparse dependence (via a biased proxy)
- CoT fine-tuning sparsifies attention (confounded)
- Expressivity is not the LLM bottleneck (an anecdote)
- Sparsity improves the landscape (§4.2)

Relation to adjacent work

Paper	Relation	How
Feng; Li; Merrill & Sabharwal	Contrasts	Asks what CoT makes learnable; Thm 1 rules out expressivity
Wies, Levine, Shashua 2023	Advances	Same claim for transformers, with feature learning, not NTK
Abbe et al. 2024 (globality)	Instantiates	CoT steps have globality 2 vs k; Thm 3 proves one instance
Raz; Kol–Raz–Tal; Lyu et al.	Changes assumptions	Brings streaming memory bounds to SGD, and their premise
Abbe & Sandon 2020	Limits	Large nets learn parity by SGD, so bounds must cap a resource
Malach 2024 (not cited)	Overlaps	Linear models learn parity from CoT data; adds a mechanism

Globality (Abbe et al.): the fewest tokens, beyond the token histogram, that correlate with the target. Without CoT it is k ; each CoT step needs only $b[i]$ and $b[\tilde{S}[i]]$.

Kim & Suzuki: same question, independent route

Axis	Wen et al. (this paper)	Kim & Suzuki (ICLR 2025)
CoT shape	chain of k steps, one secret bit per step	binary tree of 2-parities, depth $\log_2 k$
Model	frozen random embeddings, trained ReLU FFN	one-hot positions, fixed link φ
Training	3 minibatch SGD steps on fresh samples	1 GD step (teacher forcing); $\log_2 k$ (self-check)
Upper bound	$\tilde{O}(n)$ samples, n = input length	$\Omega(d^{2+\epsilon})$ samples, d = input length
Lower bound	$o(nk)$ -bit memory $\Rightarrow 2^{\Omega(k)}$ samples	approximate gradients $\Rightarrow \text{poly}(d)$ queries fail
It needs	a small model, constant passes	bounded gradient precision; any model size

Complementary: Kim & Suzuki's lower bound needs no memory cap; Wen et al.'s upper bound counts finite samples. Neither alone gives a separation for over-parameterized models trained by SGD.

CONCLUSION

What is now known, and what is not

PROVED

In a one-layer transformer with $o(nk)$ parameters, CoT supervision moves k -sparse parity from $2^{\Omega(k)}$ samples (constant-pass SGD) to $3 \cdot C_2 \cdot n \cdot \log^{20}(n/\delta)$, and the learned attention recovers S in order, within $1/n^8$.

OBSERVED

At $n = 20$, $k = 12$: about 60,000 one-pass samples with CoT; no learning in any tried configuration without it.

OPEN

Whether the separation survives over-parameterization, and whether sparse attention causes CoT's gains on real data.

POINT 8 · ONE PRECISE LIMITATION

Theorem 2 binds only learners with $o(nk)$ bits of state and $O(1)$ passes. The paper's own models exceed that budget about 10^5 -fold per layer, and many-pass training escapes it.

So the paper does not establish that CoT improves sample efficiency for over-parameterized transformers, the regime it invokes to motivate the work.

A memory sweep that decides which mechanism is real

Fix $(n, k) \in \{(30, 4), (20, 6)\}$. Sweep parameters $\times$ bits from below $nk / \ln n$ to 10^6 times above it (one layer, d from 4 to 720; 8, 16 and 32-bit). Train one pass on fresh samples, with and without CoT; record samples to 100% validation accuracy over 5 seeds.

OUTCOME A · STRENGTHENS, RE-EXPLAINS

No-CoT cost stays exponential in k far above the threshold. The gap is real but not memory-driven; the theorem to prove is a gradient-signal lower bound for any width, paired with Thm 3.

OUTCOME B · CHALLENGES

No-CoT cost collapses once state far exceeds nk bits. Thm 2's mechanism is the operative one, and CoT's sample advantage shrinks with scale.

Either result changes how the theorem should be read. The paper's sweeps never vary the one quantity its lower bound depends on.